

Quick Check

Lesson 1: Where Do Plant Wastes Go? Gum, Droplets and Falling Leaves

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 2.3: Excretion in Plants | Lesson 1

On a humid early morning, Grade 11 learners looked closely at grass and arrowroot (taro) leaves in the school garden. What did they see standing along the edges of the leaves?

- A** Yellow leaves dropping off
- B** Thick milky sap oozing out
- C** Amber resin that had hardened into a seal
- D** Small droplets of liquid

During Think-Pair-Share, a learner predicts: 'The sticky sap from a cut Acacia is the plant's blood.' At this point in the lesson, what is the best scientific way to treat this idea?

- A** Record it as a prediction and test it later using evidence
- B** Reject it at once because the teacher has not agreed with it
- C** Ignore it because predictions made before teaching do not matter
- D** Accept it as true because a classmate said it

A chart compares a running goat with a maize plant of the same body mass. It shows that the goat uses much more oxygen and food and makes much more nitrogenous waste. What does the chart show about the amount of nitrogenous waste the maize plant makes?

A The same amount as the goat

B Much less than the goat

C None at all

D More than the goat

Excretion is the removal of metabolic wastes from the body. Which of these is an example of excretion in a human?

- A** Removing urea in urine
- B** Breathing in oxygen
- C** Swallowing food
- D** Passing out undigested ugali fibre as faeces

After the debate, a learner says: 'We now have ideas about why plants have no kidneys, but we still cannot say what the droplets on the edges of arrowroot leaves are or how they form.' Which statement is correct at this stage of the unit?

- A** The plant is sick and is leaking liquid
- B** The droplets are sweat from sweat glands on the leaf
- C** How the droplets form has not been explained yet and will be investigated in the next lesson
- D** The droplets are urine made by tiny kidneys inside the leaf

Learners are writing sticky-note questions for the Driving Question Board about oozing sap, leaf-edge droplets and falling leaves. Which question best fits the board, because it is about the phenomenon and cannot be answered yet?

A Did we go on a nature walk today?

B How many leaves does one arrowroot plant have?

C What colour is the gum from an Acacia?

D Why do arrowroot leaves have water drops along their edges in the morning?

While revising her drawing, a learner puts a question mark on the arrow showing how wastes leave the plant, because she cannot explain it yet. Why is this a good scientific move?

- A** It means she did not take part in the lesson
- B** It shows a gap in her model that needs to be investigated
- C** It shows the drawing is completely wrong and must be rubbed out
- D** It means the teacher will give the answer without any investigation

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).